



Research Report for Project Seminar I

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Introduction

Boredom is a common psychological state. It can be conceptualized as an aversive state that occurs when individuals experience difficulty sustaining attention and engagement with their current activity or environment. This state involves an awareness of the lack of engagement and the perception that the current situation is responsible for this experience. Therefore, boredom can influence how people respond to their surroundings. (Eastwood et al., 2012)

Boredom is closely related to attentional engagement. According to the meaning-and-attentional-components (MAC) model, boredom can arise when individuals experience difficulties engaging their attention with an activity. This may occur when the demands of an activity are neither too low or too high relative to an individual's available cognitive resources, making it difficult to sustain attention. Attentional boredom may consequently be characterized by difficulties with concentration, mind wandering, and inattention. (Westgate, 2020). In the present study, participants experienced high- and low- boredom conditions while standing and being exposed to a virtual reality environment and a podcast.

Given that boredom is closely associated with attentional engagement, it is relevant to consider whether changes in attention might also influence other processes, such as postural control. According to (Vuillerme, 2005), postural sway can be influenced by cognitive and attentional processes. Examining specific parameters may provide additional insight into how different boredom conditions affect postural control.

Postural sway can be divided into different components. According to (van der Kooij & de Vlugt, 2007) there are periodic and remnant components of postural responses. While the periodic component reflects responses associated with external perturbations, the remnant component represents the non-periodic part of the sway signal. Because remnant power captures postural variations that are not directly explained by external stimulus, it was selected as the outcome measure in the present study.

Although boredom has been studied extensively in relation to attention and cognitive engagement, less is known about whether different levels of boredom are associated with differences in spontaneous/non-periodic components of postural sway. Therefore, the research question for this study is: Does remnant power differ between high- and low-boredom conditions?

Methods

In the present study, participants completed balance measurements under two boredom conditions: high boredom (HB) and low boredom (LB). During the experiment, participants stood in an upright position while wearing a virtual reality headset and listening to podcast material, while body sway was measured. Demographic and psychological information was additionally collected using questionnaires.

Participants

The final sample size for the experiment was 41 participants (27 participants in 2024 and 14 participants in 2026). Demographic information, including age, height, and weight, was collected and included in the dataset. Requirements for participating in the study were to be healthy and not have epilepsy.

Experimental Design and Procedure

The experiment was conducted partly in the summer 2024 and partly in the summer 2026 in the HPRC (Human Performance Research Centre) in University of Konstanz and it was completed in a laboratory setting. Upon arrival, participants received information about the study and completed an informed consent form and a safety questionnaire. To avoid influencing participants' expectations regarding the experimental manipulation, the study was introduced as an investigation of balance and virtual reality rather than boredom. Participants' height and weight were measured, and they completed a baseline LimeSurvey questionnaire before the experimental measurements began.

For the measurements, participants removed their shoes, jewellery, and items from their pockets. Tracking markers were attached to the back, wrists, and hip, and participants were equipped with a VR headset and headphones. The VR headset was individually calibrated and the audio volume adjusted to a comfortable level. Participants were also given a handheld clicker and instructed to count occurrences of a predefined target word in the audio material. They were informed about the procedure and reminded that they could discontinue the experiment at any time.

The experiment consisted of a high-boredom (HB) and a low-boredom (LB) condition, with the order of conditions counterbalanced across participants. In the LB condition, participants selected a podcast from a range of interesting topics. In the HB condition, participants listened to podcast material about stairs, which was intended to induce a higher level of boredom. Each condition consisted of four measurement trials (t1–t4). During the trials, participants stood in the designated measurement position while wearing the VR headset and listening to the assigned audio material. Balance/VR measurements and the PsychoPy task were recorded simultaneously. Between trials, participants responded to check-up questions, and the cumulative clicker count was recorded.

After the first condition, participants removed the VR headset and completed a LimeSurvey questionnaire. This was followed by a 5–8-minute break during which participants were free to move around. Before the second condition, participants completed the relevant LimeSurvey questions and the measurement equipment was checked and recalibrated. The second condition was then conducted following the same measurement procedure as the first.

After completing the second condition, participants completed a final LimeSurvey questionnaire. The equipment was removed, participants were debriefed and thanked for their participation, and observations from the session were documented. The recorded data were subsequently checked for obvious recording problems and correct participant identification before storage.

Materials and Measures

Postural sway was recorded using a lighthouse-based tracking system. Tracking markers were attached to the participant's back, wrists, and hip, allowing body movement to be recorded while participants stood upright during the experimental trials. Participants wore a virtual reality headset and headphones throughout the measurements.

Auditory material was presented through the headphones. In the low-boredom condition, participants selected from a range of interesting podcasts, whereas the high-boredom condition used podcast material about stairs. A handheld clicker was used to count occurrences of a predefined target word in the audio material.

Demographic and psychological information was collected using LimeSurvey. This included participant characteristics as well as repeated self-report measures during the experimental procedure, including boredom ratings.

The primary outcome measure for the present study was Remnant Power, representing the non-periodic component of postural sway as described in the Introduction. Remnant Power was obtained for four trials (t1–t4) in each boredom condition and was subsequently used to compare postural sway between the high- and low-boredom conditions.

Data processing and analysis

Data were imported and processed in R using Positron. The complete dataset was preprocessed before variables relevant to the research question were selected for the final analysis. The analysis was conducted in R, and version control was managed using Git and GitHub. Raw data files were not manually modified during data processing.

Remnant Power variables from the 2024 and 2026 datasets were converted to a consistent numeric format before the datasets were combined. For each participant, mean Remnant Power was calculated separately for the high-boredom (HB) and low-boredom (LB) conditions across the four trials

(t1–t4). For participants with partially missing trials, condition means were calculated using the available trials.

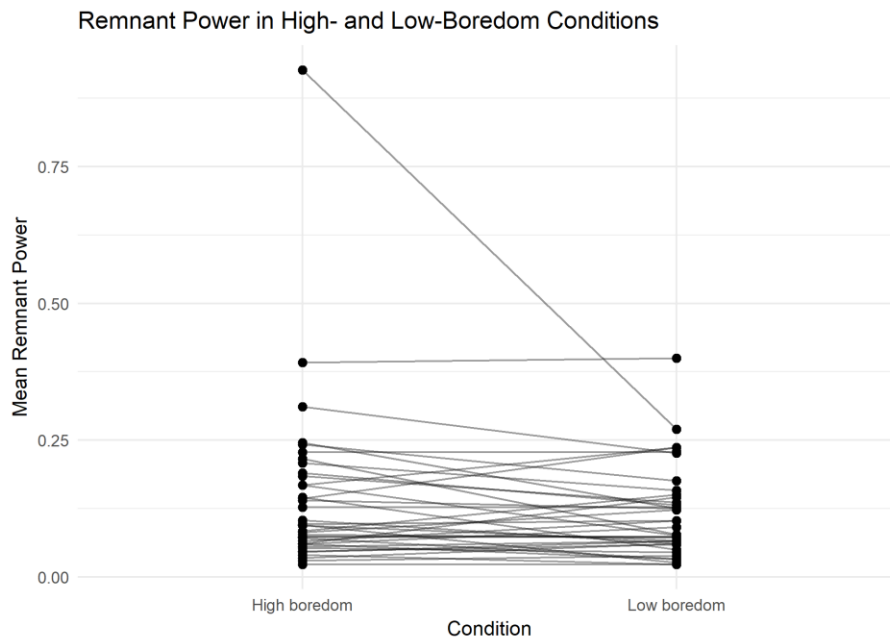
Remnant Power was available for 41 participants across the 2024 and 2026 datasets. Three participants were excluded from the paired high- versus low-boredom analysis because Remnant Power data were completely missing for one experimental condition, leaving 38 participants for analysis. No additional participants were excluded based on outliers.

The distribution of the within-participant differences between HB and LB Remnant Power was examined using descriptive statistics, a histogram, and a Q-Q plot. As the differences showed a clear deviation from normality, a paired Wilcoxon signed-rank test was used to compare Remnant Power between the two conditions. Descriptive statistics were reported as medians and interquartile ranges (IQRs).

Results

The final analysis included 38 participants. Median Remnant Power was 0.0942 (IQR = 0.120) in the high-boredom condition and 0.0841 (IQR = 0.0825) in the low-boredom condition. A paired Wilcoxon signed-rank test showed no statistically significant difference in Remnant Power between conditions, $V = 462$, $p = .187$.

Figure 1. Mean Remnant Power for individual participants in the high-boredom (HB) and low-boredom (LB) conditions.



Note. Lines connect measurements from the same participant.

Discussion

The present study investigated whether Remnant Power differed between high- and low-boredom conditions. Although median Remnant Power was slightly higher in the high-boredom condition than in the low-boredom condition, the difference was not statistically significant. Thus, the present findings do not provide evidence that Remnant Power differed between the two boredom conditions. The slightly higher median Remnant Power observed during high boredom may suggest greater non-periodic postural sway in this condition; however, the statistical analysis does not support interpreting this difference as a reliable effect.

Several limitations should be considered when interpreting these findings. First, the final analysis included only 38 participants, which may have limited the ability to detect small differences in Remnant Power between the two boredom conditions. Missing data also reduced the available

information. Three of the 41 participants were excluded because Remnant Power data were completely missing for one condition, and some of the remaining participants had individual trials with missing values. For these participants, condition means were therefore based on fewer than four trials. In addition, boredom manipulation may not have affected all participants equally. While participants in the low-boredom condition could choose from a selection of interesting podcasts, the high-boredom condition consisted of podcasts about stairs. As boredom is subjective, material intended to be boring or interesting may not necessarily produce the same level of boredom in every participant. These factors should be taken into account when interpreting the absence of a statistically significant difference between conditions.

In conclusion, the present study found no statistically significant difference in Remnant Power between high- and low-boredom conditions. Although Remnant Power was descriptively higher during high boredom, the observed difference was not statistically reliable. Further research with larger samples and robust manipulation checks could help clarify whether boredom influences the non-periodic component of postural sway.

References

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